

Collaborative Learning Discussion Summary

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Module: Security and Risk Management

1. Joining the Discussion

I have participated actively in the collaborative discussion forums to synthesize risk management concepts throughout this module. I focused on the practical implementation of vulnerability management frameworks, specifically moving from subjective scoring systems (CVSS) to more resilient, context-aware decision frameworks (SSVC).

2. Major Contributions Summary

In my main post I had a critical look at the limitations of CVSS and pointed out that subjective interpretation frequently results in inconsistent risk scoring (Wunder et al., 2024). Peer engagement helped me shift my perspective from one of a single-metric focus to a pragmatic, hybrid approach:

- **The Hybrid Strategy:** I suggested merging CVSS (for technical severity) + EPSS (for attack likelihood) with SSVC decision trees to evaluate risks to physical hardware end-points.
- **Communication Value:** I realized that although CVSS is flawed for prioritization, it is a “universal language” for communication with external third-party vendors.

3. Responses to Peers

I collaborated with my peers to talk about practical solutions for resource misallocation caused by misleading vulnerability scores:

- **Advocacy for SSVC:** I argued that implementing Stakeholder-Specific Vulnerability Categorization (SSVC) effectively eliminates the “alert fatigue” that artificial CVSS scores create (Spring et al., 2021).
- **Asset-Centric Risk Mapping:** I advocated for mandatory contextual risk-mapping based on well established asset inventories in my peer responses, rather than vague metrics like “Scope” or “User Interaction” (Wunder et al., 2024). Meaning defensive resources are targeted at real-life threats, not subjective assumptions.

4. Reference

Aven, T. and Thekdi, S. (2025) Risk Science: A Primer. 2nd edition. Oxon: Routledge.

Spring, J. M., Hatleback, E., Householder, A. D., Manion, A. and Shick, D. (2021)
Prioritizing response to vulnerability: A stakeholder-specific categorization of vulnerability (version 2.0), Software Engineering Institute, Carnegie Mellon University.

Wunder, J., Kurtz, A., Eichenmüller, C., Gassmann, F. and Benenson, Z. (2024)
Shedding light on cvss scoring inconsistencies: A user-centric study on evaluating widespread security vulnerabilities. *IEEE Symposium on Security and Privacy (SP)*, pp. 58-75.